Solution

Winter End-Semester Examination- 2023-24

Seismological Data Analysis (GPD508)

Ans 1a. The ambient seismic wave field, also known as ambient noise, is excited by oceanic gravity waves primarily. This can be categorized as seismic hum (1–20 mHz), primary microseisms (0.02–0.1 Hz), and secondary microseisms (0.1–1 Hz). Below 20 mHz, pressure fluctuations of ocean infragravity waves reach the abyssal floor.

Ans 1b. Discussed in the class. See the class notes.

Ans 1c. The receiver function technique is a useful tool to understand about seismic discontinuities in the Earth. SRFs have higher wavelength than PRFs and can penetrate deeper with .

Ans 2a cat /etc/shells

Ans 2b. script.sh &

Ans 2c. With set, you can turn on or off debugging options in the shell:

Set -x: This displays commands and their arguments as they are being executed.

Set -v: It displays shell input lines in real-time as they are read.

Ans 2d. The following commands are available to check disk space usage:

- df – It is used to find out how much space is left on a disk.
- du – With this command, you can find out how much disk space the specified files and each subdirectory take up.
- dfspace – Using this command, you can check the amount of free disk space in MB

Ans 3a.

taup time - calculates travel times.

taup pierce - calculates pierce points at model discontinuities and specified depths.

taup path - calculates ray paths, depth versus epicentral distance.

taup table - outputs travel times for a range of depths and distances in an ASCII file.

taup create creates a .taup model from a velocity model.

Ans 3b.

```
#!/bin/bash
MyFile=cars.txt
if [ -f "$MyFile" ]; then
echo "$MyFile exists."
```

```
else
echo "$MyFile does not exist."
fi
```

Ans 3c.

- **(i)** Change to end of line
- **(ii)** Edit new file
- **(iii)** The following example replaces **every** occurrence (caused by the `g` at the end of the command) of `apple` with `pear.`
- **(iv)** `Delete n words`
- (v) Write lines 30 to 60 in file new

Ans 3d.

- The Vp/Vs ratio is sensitive to composition, heat flow, and fluid fraction of the crust (Christensen, 1996).
- For common igneous rocks, Vp/Vs values are ~1.73 for felsic-quartz granite, ~1.78 for intermediate compositions, such as diorite, and ~1.88 for mafic rock, such as basalt or gabbro.
- The presence of partial melt or fluids serves to raise Vp/Vs ratios (Watanable, 1993; Thomson et al., 2010). Note that Vp/Vs can be uniquely converted to Poisson ratio, which is a more common parameter in the Earth sciences.

Ans 4a.

We know that

$V_R = 0.92\ V_s$,

Here, V_R and V_s are the velocity of the rayleigh wave and shear wave, respectively.

Given: V_s = 3.25 km/s

Put value of V_s in above equation,

V_R = 0.92*3.25 = 2.99 km/s ~ 3.0 km/s

Ans 4b.

If the Wavelength of the tsunamis waves is significantly longer than the ocean floor depth, the phase and group velocity are the same. So it is non-dispersive, and we can calculate the phase velocity using the formula V = gd

Given: d = 9 km; g = 9.8 m/s^2 = 0.0098 km/s^2

Put these values in above equation

V=0.0098*9=0.296 km/s ~0.3km/s

Ans 4c. Discussed in the class

Ans 5.

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Rayleigh waves in homogeneous medium

* Rayleigh waves are combination of P & SV waves that can exist at the top of homogeneous half space $z=0$ [free surface]

Particle motion of Rayleigh wave is in x-z plane, we consider only P & SV waves, bcz they can satisfy the free surface boundary condition & don't interact with SH waves. The P & SV potential are:

$$\phi = A \exp\left[i(\omega t - k_x x - k_z z)\right]$$

$$\psi = B \exp\left[i(\omega t - k_x x - k_z z)\right]$$

To describe energy trapped near the free surface, two conditions must apply. The soln must both ensure that the energy must not propagate away from surface & satisfy free surface boundary conditions:

1. For this purpose, $e^{-k_z z} \longrightarrow 0$ if $z \to \infty$
2. Traction at free surface must vanish.

To solve potential wave eqn using above two condn:

$$c_x = \left(2 - \frac{2}{\sqrt{3}}\right)\beta = \underbrace{0.92\beta}$$

Apparent vel. of Rayleigh wave along free surface

Shear wave vel.

Apparent vel. of Rayleigh wave along free surface is
... lesser than

At $z=0$, the displacement components are

$$u_x = 0.42\, A k_x \sin(\omega t - k x)$$
$$u_z = 0.62\, A k_x \cos(\omega t - k_x x)$$

Poisson's Solid

To visualize the above two eq^ns at $z=0$; consider there is a particle at $x=0$

At $t=0$, u_z is a maximum & $u_x=0$.

As time increases the x & z displacements combine to give counterclockwise or "retrograde" elliptical motion.

For Poisson Solid, the maximum vertical displacement at the surface is about 1.5 times the maximum horizontal displacement.

An

slightly lesser than Shear wave velocity.

The coeff. of potential are related as

$$B = A\left(2 - \frac{c_x^2}{\beta^2}\right)/2r_\beta \quad \text{where} \quad r_\beta = \left(\frac{c_x^2}{\beta^2} - 1\right)^{1/2}$$

Relationship b/w potentials & displacement are as follows:

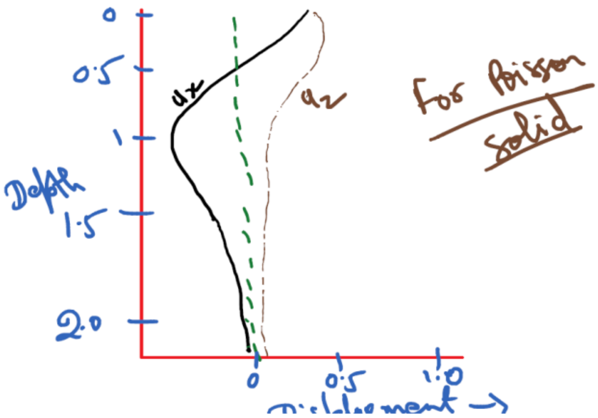
$$u_x = \frac{\partial\phi}{\partial x} - \frac{\partial\psi}{\partial z} \quad \& \quad u_z = \frac{\partial\phi}{\partial z} + \frac{\partial\psi}{\partial x}$$

So, for Poisson Solid

$$u_x = A k_x \sin(\omega t - k_x x)\left[\exp(-0.85 k_x z) - 0.58\exp(-0.39 k_x z)\right]$$

$$u_z = A k_x \cos(\omega t - k_x x)\left[-0.85\exp(-0.85 k_x z) + 1.47\exp(-0.39 k_x z)\right]$$

Both displacements decays with depth w.r.t $e^{-k_x z}$, so the depth to which a Rayleigh has significant displacement is proportional to its horizontal wavelength.



Ans 5. Discussed in the class.

Ans 6. Discussed in the class.

Ans 7. Discussed in the class.